

BRAIN TUMOR CLASSIFICATION USING DEEP LEARNING

**A CONVOLUTIONAL NEURAL NETWORK APPROACH FOR
MULTI-CLASS MRI CLASSIFICATION**

PRESENTED BY:

UCE2023518 - AMRUTA DESHMUKH

UCE2023525 - TANVI GAGAN

UCE2023538 - SMRUTI MODAK

UCE2023547 - RUCHIKA PATIL

FINAL MODEL ACCURACY (TEST SET): 98%

THE PROBLEM:

- Manual analysis of brain MRIs is a specialized, time-consuming, and subjective task.
- Early and accurate diagnosis is critical for patient outcomes.

PROJECT OBJECTIVES:

- To develop a model to automatically classify brain MRIs.
- To accurately distinguish between four categories: Glioma, Meningioma, Pituitary Tumor, and No Tumor.
- To build and validate a high-performance Convolutional Neural Network (CNN) for this task.

DATASET: THE FOUNDATION OF OUR MODEL

- Over 7,023 total MRI images 4569 trained contrast-enhanced MRI images.
- Validated: 1143 and Tested: 1311 Images across four classes.

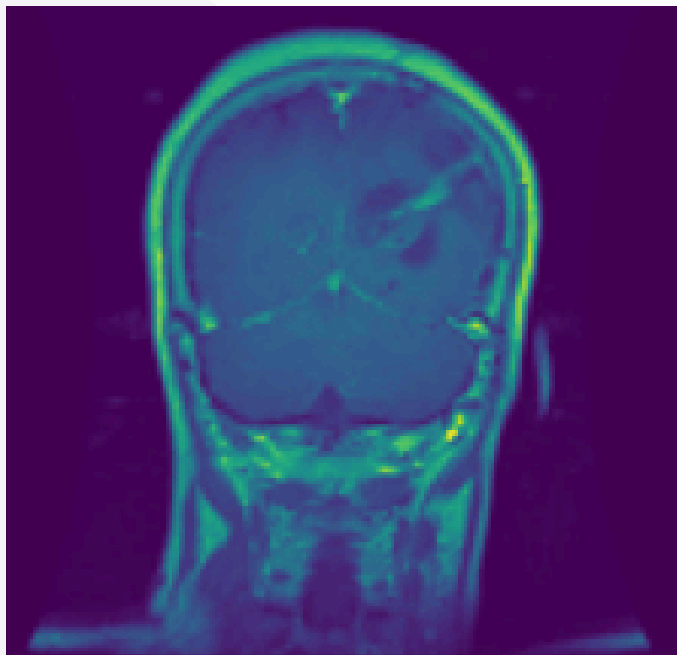
[DATASET LINK](#)

Tools and Technologies:

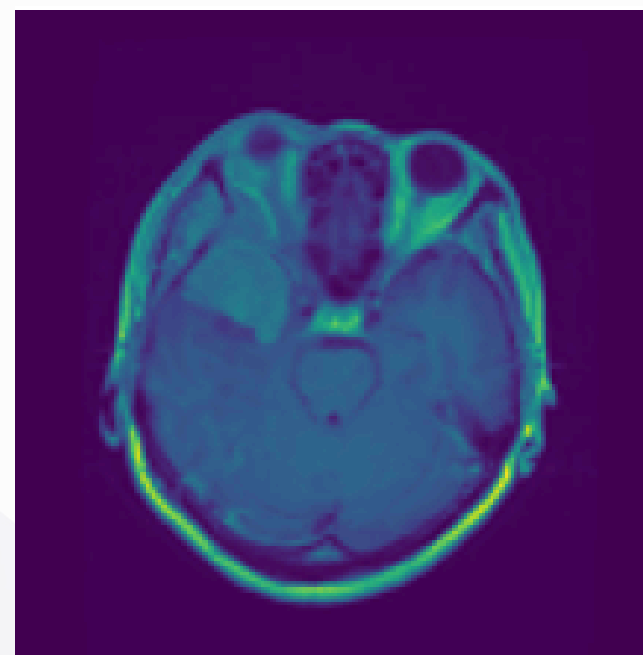
- **Deep Learning: TensorFlow/Keras** for model building.
- **Model: VGG16** (Transfer Learning) for feature extraction.
- **Environment: Google Colab** for development and GPU access.
- **Language:** Python for all scripting.
- **Data/Metrics: NumPy, Pandas, Scikit-learn** for numerical handling, data processing, and evaluation metrics (Confusion Matrix, ROC/AUC).
- **Visualization:** Matplotlib/Seaborn for generating all graphs.

EXPLORATORY DATA ANALYSIS: VISUALIZING THE DATA

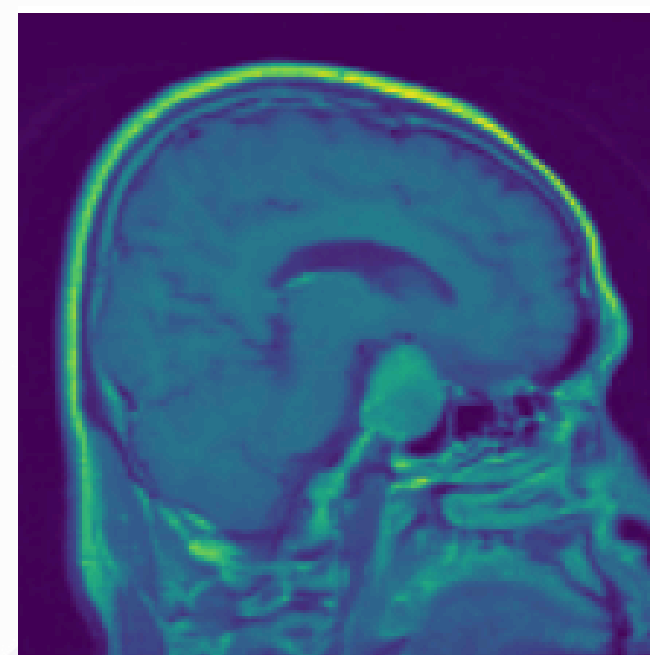
A VISUAL INSPECTION OF THE FOUR DISTINCT CLASSES.



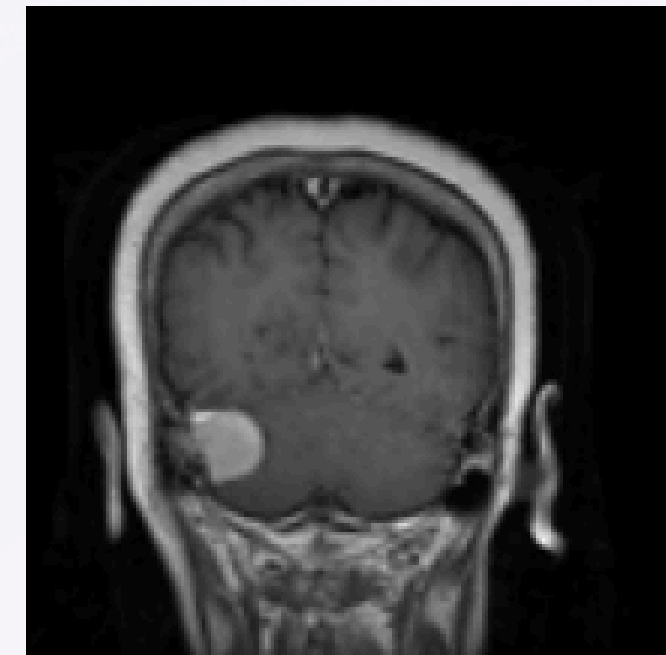
GLIOMA



MENINGIOMA



PITUITARY



NO TUMOR

METHODOLOGIES

Pre-processing:

1. All images resized to 224x224 pixels (VGG16 standard input size).
2. Pixel values normalized from [0-255] to [0-1] range.

Augmentation (Critical for medical data): Used random brightness/contrast adjustments during training to prevent overfitting and enhance generalization.

Architecture:

1. VGG16 Base: Loaded weights trained on the massive ImageNet dataset.
2. Fine-Tuning: The last four convolutional layers of VGG16 were unfrozen and trained, allowing the model to adapt general features to specific MRI patterns.
3. Custom Head: Added layers for classification: Global Average Pooling, Dropout (0.5), Dense (256), Batch Normalization, and a Final Dense (4 neurons) layer with Softmax activation.

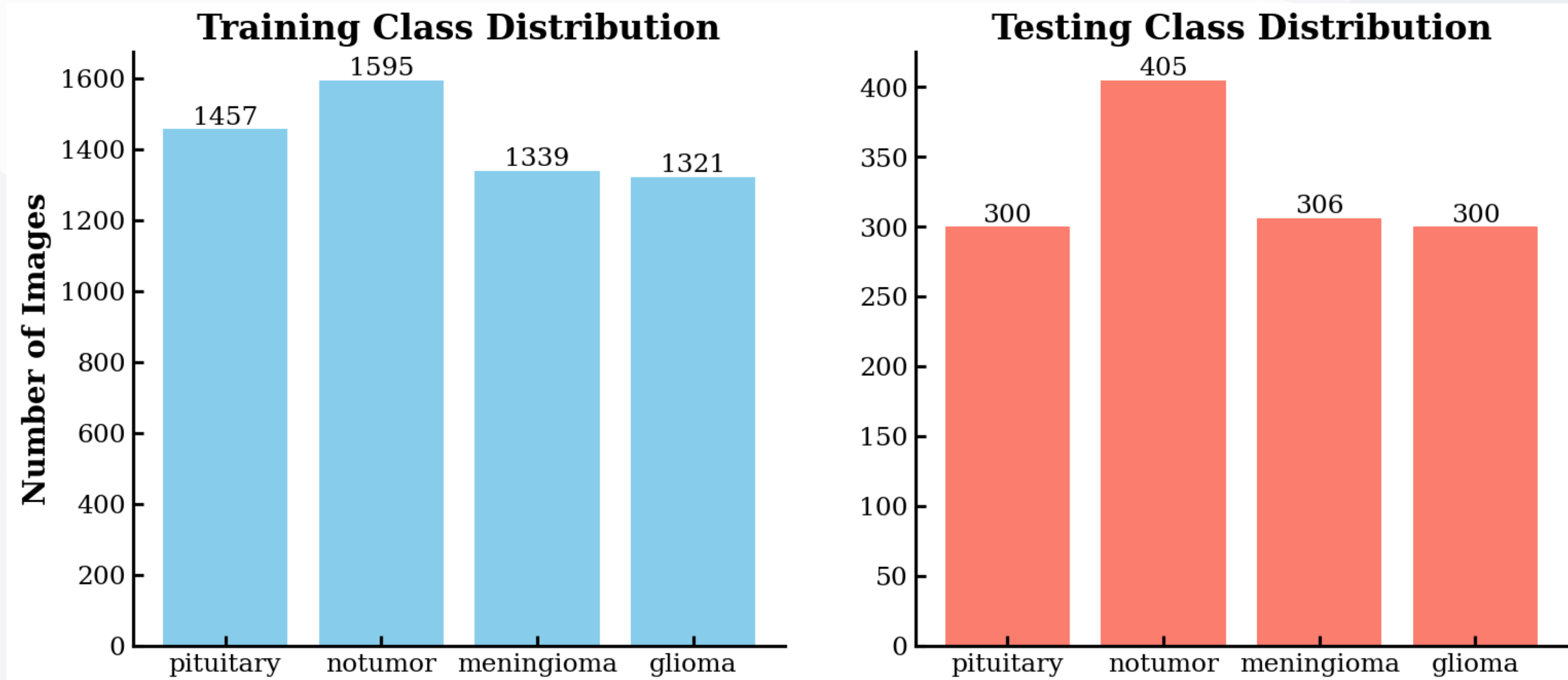
TRAINING PARAMETERS

KEY PARAMETERS:

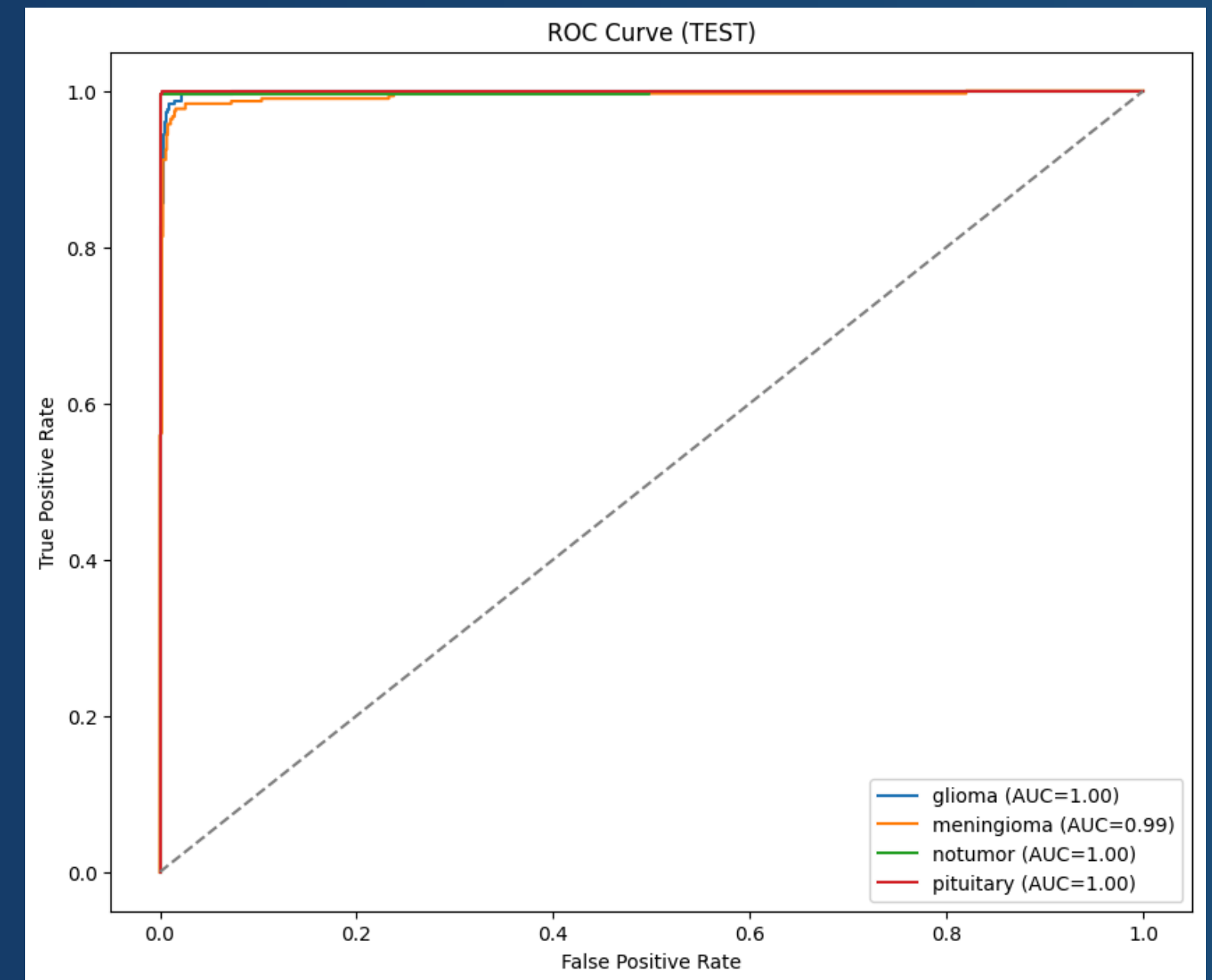
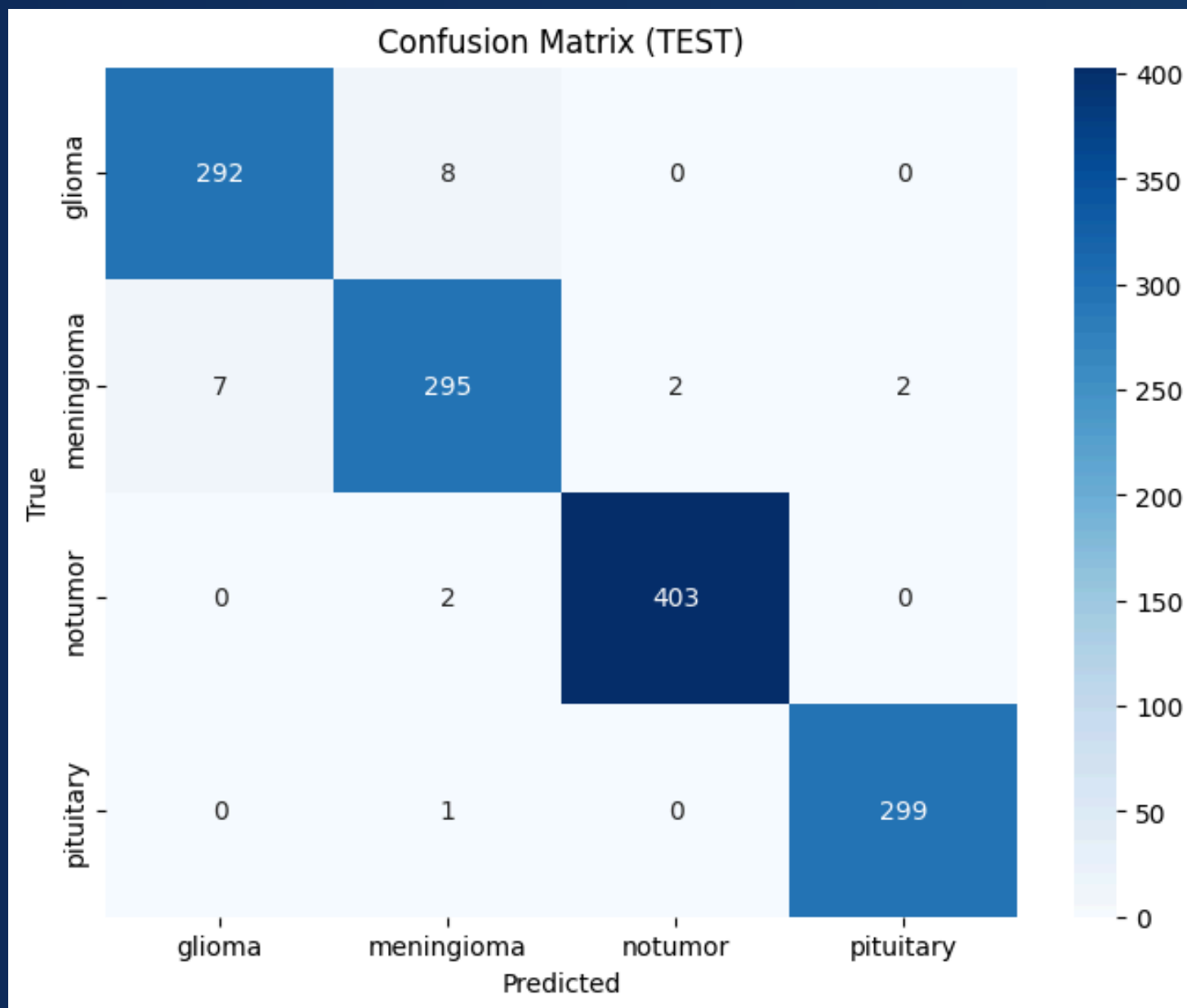
- **Optimizer:** Adam (Learning Rate: 0.0001)
- **Loss Function:** Sparse Categorical Crossentropy
- **Batch Size:** 32
- **Epochs:** 25 (with Early Stopping)
- Both training and validation loss converged to near-zero values, confirming the model successfully learned the distinction between the four classes.
- **Regularization:** Used **Dropout (0.5 and 0.3)** and **Batch Normalization** to ensure the model generalizes well and avoids overfitting.
- **Callbacks:** **Early Stopping** (patience 10) and **ReduceLROnPlateau** were used to optimize training time and stability.

EXPLORATORY DATA ANALYSIS: CLASS BALANCE

THE BAR CHART SHOWING THE COUNT FOR EACH OF THE 4 CLASSES.



ACCURACY& ERROR MINIMIZATION



FINAL PERFORMANCE ON THE TEST SET

- Overall Metrics:
- Overall Accuracy: 0.98 (98%)
- Macro/Weighted Average F1-Score: 0.98
- Per-Class Strength: The model achieved 1.00 Precision/Recall for 'notumor' and close to perfect scores for the other three tumor types.

Classification Report (TEST):

	precision	recall	f1-score	support
glioma	0.98	0.97	0.97	300
meningioma	0.96	0.96	0.96	306
notumor	1.00	1.00	1.00	405
pituitary	0.99	1.00	1.00	300
accuracy			0.98	1311
macro avg	0.98	0.98	0.98	1311
weighted avg	0.98	0.98	0.98	1311

OBSERVATIONS AND RESULTS

- **Observations:** The 22 misclassified images often show complex, non-standard tumor shapes or poor image contrast.
- **Classification & ANN:** Successfully developed and evaluated a deep learning classification model (**VGG16 CNN**).
- **Pre-processing:** Implemented data normalization and augmentation for high-quality input.
- **Application & Evaluation:** Built a robust ML application, achieving a high-level outcome on a real-world problem.

APPLICATION & GUI: REAL-TIME DIAGNOSTIC SUPPORT

- **Deployment Platform:** The final model is deployed using **Streamlit**, which creates an interactive, user-friendly web application (GUI).
- **The Model's Role:** The app loads the trained **VGG16 model** (final_clean.keras) to serve as the core prediction engine.
- **User Interaction:** The user simply **uploads an MRI image** via the web interface.
- **Real-Time Processing:** The system automatically performs necessary pre-processing (resizing to **224x224** and normalization) before feeding the image to the model.
- **Result Output:** The GUI instantly displays the **Predicted Tumor Class** (e.g., glioma), the model's **Confidence Score** (e.g., 99.2%), and the full probability breakdown for all four tumor types.
- **Practical Use:** This application functions as an effective **Clinical Decision Support System**, providing doctors with a fast, objective "second opinion" to enhance diagnostic accuracy and speed.

CONCLUSION & LIMITATIONS

- **Conclusion:** The VGG16 Transfer Learning model achieved state-of-the-art performance with 98% overall accuracy in multi-class brain tumor detection.
- **Limitations:**
 1. Model performance is dependent on the quality of MRI scans.
 2. The model performs classification but does not provide localization (segmentation) of the tumor, which is a next logical step.

FUTURE ENHANCEMENTS

- **Technical Future Work:**

1. Integration of Segmentation: Modify the model to perform instance segmentation (identifying the exact boundaries of the tumor).
2. Alternative Architectures: Test newer models like EfficientNet or ResNet to push the accuracy to 99%+.

- **Deployment:** Finalize the Flask UI deployment for clinical demonstration and testing.

**THANK
YOU.**

